Journal of Veterinary Behavior 88 (2026) 85–98

Contents lists available at ScienceDirect

Journal of Veterinary Behavior

journal homepage: www.elsevier.com/locate/jveb

Caregiver bereavement-related distress and surviving dog behavior after pet dog loss: Associations with euthanasia and expected death in a cross-sectional study from Italy and the UK

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ARTICLE INFO

Keywords:
Dog behavior
Euthanasia
Pet loss
Grief

ABSTRACT

This exploratory cross-sectional study investigated whether, and how, the circumstance of a dog's death (euthanasia vs. unassisted) and whether this death was expected were associated with caregivers' grieving processes and reported behavioral changes in a cohabiting dog in Italy and the UK. The Mourning Dog Questionnaire (MDQ), which combines questions on caregivers, dogs, death circumstances, and surviving-dog behavior with five validated psychometric scales (Pet Bereavement Questionnaire - PBQ, Lexington Attachment to Pets Scale - LAPS, Animal–Human Continuity Scale - AHCS, Positivity Scale - P-Scale, and Testoni Death Representation Scale - TDRS), was completed online by 655 caregivers (65% from Italy; 35% from the UK). Multiple regression analyses showed that both euthanasia and expected death significantly predicted reduced caregiver grief distress across the dimensions of grief, anger, and guilt. Higher caregiver grief and anger were associated with an increased likelihood of the surviving dog showing behavioral changes such as diminished appetite, increased vocalization, fearfulness, attention-seeking, avoidance of the deceased dog's items, and prolonged alterations. UK participants reported significantly lower guilt but higher grief than Italians and were more likely to report some canine behavioral changes, such as sleeping more and longer persistence of alterations. These findings highlight the importance of integrating culturally sensitive caregiver support into veterinary end-of-life care, not only to address human bereavement but also to promote the emotional adjustment of surviving dogs.

Introduction

According to recent estimates, there are over 106 million domestic dogs in Europe (Annual report | FEDIAF, 2026), with 31% of UK households (McMillan et al., 2024) and 25% of Italian households owning at least one dog (Pirrone et al., 2015a, 2015b). Multi-dog households are particularly common, accounting for approximately one quarter of dog-owning households in the UK (PDSA, 2026) and more than 15% in Italy (Assalco Zoomark ASSALCO, 2025).

Given the increasing longevity of companion dogs and advances in

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https://doi.org/10.1016/j.jveb.2026.10.005
Received 3 March 2026; Received in revised form 24 August 2026; Accepted 18 October 2026
Available online 21 October 2026
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veterinary care (Sexton et al., 2024), many caregivers and dogs are likely to experience the loss of a canine companion during their lifetime. Scientific research supports the concern that the death of a pet dog, often a companion with whom the caregiver had formed a strong emotional bond and whom they considered a family member (Uccheddu et al., 2019), can constitute a major life event, potentially leading to a decline in the bereaved person's physical and emotional health (Davis, 2011; Cordaro, 2012; Kemp et al., 2016; Cowling et al., 2020). However, the effects of a dog's death on caregivers are still poorly investigated (Brown et al., 2023). Relatively little is known about the factors that may help predict how individuals experience and go through their grieving process. Reprocessed death of a companion animal can indeed vary widely across individuals (McCutcheon and Fleming, 2002; Cowles, 2018), and emerging evidence suggests that this variation might be influenced by several contextual factors. For example, (Uccheddu et al., 2019) an unexpected death was often associated with intense feelings of grief and anger among Italian dog caregivers, possibly due to its suddenness or the absence of adequate preparation and support, such as that provided by a veterinarian (Uccheddu et al., 2019).

The modality of death, whether accidental, natural, or through euthanasia, also appears to influence bereavement (McCutcheon and Fleming, 2002; Stokes et al., 2002), although findings are mixed. Euthanasia is a leading cause of death in domestic dogs (Rajaram et al., 1993; Stokes et al., 2002) in certain countries, including the USA, the UK and Italy, regardless of whether it is prompted by illness, age, behavioral issues or other factors. While some caregivers of euthanized animals experience intense grief (Testoni et al., 2017), often burdened by doubts or regret about whether their pet might have recovered, others report a sense of comfort in having provided their pet with a dignified death (Rajaram et al., 1993) by means of this widely accepted practice (Testoni et al., 2017). This perception may be reflected in findings from our previous study by (Uccheddu et al. (2019), which showed that a dog's euthanasia was significantly associated with lower levels of caregiver anger and guilt as compared to when the dog died unassisted. Additionally, some caregivers entirely refuse euthanasia on ethical grounds, believing it is wrong to act to end another living being's life (Testoni et al., 2017). The option of euthanasia can thus be emotionally complex for people (McCutcheon and Fleming, 2002), offering relief, emotional strain, or often a combination of both (Vonk, 2025; Vonk et al., 2022).

This variability is of particular importance, not only because recognizing and understanding it can help guide appropriate support strategies for caregivers, but also because the emotional responses of grieving caregivers might influence the behavioral and emotional wellbeing of other dogs in the household (Uccheddu et al., 2022). In fact, according to caregivers' reports, from our previous study surviving dogs exhibited observable changes in activity patterns (e.g., decreased "play", "sleep", and "eating") and emotions (increased fearfulness) when a close canine companion died, with aspects of the latter (emotional dimension) being linked to the caregiver's own emotional status. Similarly, Walker et al., (2016) found that, after the loss of an animal companion, dogs were commonly reported to show changes in affectionate and territorial behaviors, including increased attention-seeking and seeking out the deceased animal's favorite places, as well as changes in food consumption and sleeping. Yet while scientific literature, albeit still scarce and disparate, has begun to document the emotional toll of pet loss on humans, far less is known about how such loss may affect other cohabiting animals including dogs (Walker et al., 2016; Uccheddu et al., 2022; Park et al., 2023; Greene and Vonk, 2024), despite the importance of this in terms of animal welfare.

Building on our previous works (Uccheddu et al., 2019, 2022), the present study represents a further step within a broader research program based on the Mourning Dog Questionnaire. In previous work, we first examined caregiver grief-related responses and associated psychological dimensions in an Italian subsample and subsequently investigated behavioral changes in surviving dogs and their associations with caregiver grief in the same population. Building on these findings, the present study focuses on an Italy–UK subsample to examine whether two death-related contextual variables, euthanasia and expected death, are associated with caregiver psychological responses and, directly or indirectly, with reported behavioral changes in the surviving dog.

The psychological constructs examined in this study were those derived from the multidimensional framework of the MDQ. These constructs were selected on the premise that bereavement following the loss of a companion dog cannot be adequately characterized by grief intensity alone. In addition to bereavement-specific distress, expressed through grief, anger, and guilt, the experience of loss may be shaped by the strength and nature of the bond with the deceased dog, the extent to which the dog is perceived as comparable to a human family member, the caregiver's broader positive orientation toward life, and the meaning assigned to death. Accordingly, the MDQ integrates measures of pet bereavement distress, attachment, animal–human continuity, positivity, and death representation, thereby capturing complementary emotional, relational, attitudinal, and cognitive dimensions of pet loss. These dimensions were therefore retained because they are theoretically relevant to pet bereavement and had been used in the previous stages of this research program.

Using the MDQ, here we aimed to assess whether caregiver psychological responses and reported behavioral and emotional changes in surviving dogs were associated with: (1) the method of death, namely euthanasia versus unassisted death; and (2) whether the death was expected or unexpected, as reported by the caregiver. The study was conducted in two culturally distinct Western countries, Italy and the UK, which differ in euthanasia-related practices and regulations: in Italy, euthanasia is legally restricted to exceptional cases, such as incurable illness or proven dangerousness, whereas in the UK it is permitted in a wider range of circumstances, including for unclaimed strays. Based on previous findings, we hypothesized that euthanasia and expected death would be associated with lower levels of caregiver grieving, particularly in relation to anger and guilt. We further hypothesized that these caregiver grief-related responses would predict reported behavioral and emotional responses in the surviving dog, potentially reflecting more effective coping. In contrast, we did not expect that these death-related circumstances might be directly linked to behavioral and emotional changes in the surviving dog. A direct association would imply that dogs responded differently depending on whether the companion's death was assisted or unassisted, or expected or unexpected; however, while, on the one hand, to our knowledge, no prior research has investigated this link, on the other hand, dogs' ability to understand these contextual aspects of death has not yet been empirically demonstrated (Dickinson and Hoffmann, 2016; Monsó, 2019). Such a result would lay the foundation for more rigorous empirical testing of this intriguing hypothesis. Therefore, any associations between death-related circumstances and surviving dog behavior were expected to be more likely indirect, through caregiver emotional responses and the post-loss household environment, rather than direct.

Materials and methods

Procedure

Participants were recruited in October 2018 from a cross-sectional convenience sample of Italian and UK pet dogs' caregivers using social media and word of mouth. The present dataset was drawn from the same broader MDQ-based research program as our previous publications. Data from the Italian subsample had been included in earlier analyses focused on the development of the MDQ and on caregiver grief and surviving dog behavior in the Italian sample (Uccheddu et al., 2019, 2022). The present study combines the Italian and UK subsamples and addresses a distinct research question, namely the associations of euthanasia and expected death with caregiver grief-related responses and surviving dog behavior. Caregivers were required to report the date

when their companion dog passed away. The study received ethical approval from the University of Padova Ethics Committee (46DF1164A03D63129CEE38D7571F8FB7). All procedures were conducted in adherence with the relevant guidelines and regulations of our institutions. Participants received written information about the aim of the study, the procedures involved, and their right to withdraw at any time. They were also informed that the survey was part of a research project exploring the experiences of both caregivers and their dogs following the loss of a companion dog. Anonymity and confidentiality of the questionnaire were assured. Written informed consent was obtained from each participant prior to data collection, and participation was entirely voluntary.

Participants

To be eligible, participants were required to be aged 18 or above, to have experienced the loss of a dog, and to have had at least two dogs at the time of the deceased dog's passing.

Material

Instruments

In the current study, the MDQ, developed by Uccheddu et al. (2019) was administered via the online platform Google Forms.

The questionnaire consisted of two sections (see Uccheddu et al., 2022, 2019) for a detailed description). Briefly, the first section collected demographic information about the caregiver and the deceased dog, details of their relationship, and the time and circumstances of the dog's death. It also assessed the nature of the relationship between the two cohabiting dogs, shared resources and activities prior to the loss, whether the surviving dog had seen the body of the deceased dog, and behavioral and emotional changes observed in the surviving dog following the death. The relationship between the two dogs was assessed across four dimensions, namely friendly, agonistic, mutually tolerant, and parental, each rated on a 5-point Likert scale ranging from strongly disagree to strongly agree. Shared resources, including food, resting areas, and objects/toys, were coded as present or absent, while shared activities, including sleeping, playing, grooming each other, and fighting, were rated on a 3-point Likert scale ranging from never to often. Behavioral and emotional changes in the surviving dog were assessed across several domains, including playing, sleeping, eating, fearfulness, vocalization, elimination, attention-seeking, activity level, and other behaviors. For each domain, caregivers indicated whether the behavior had increased, decreased, remained unchanged, or whether they did not know. When a change was reported, caregivers were also asked to indicate its duration.

The second section comprised a combination of five validated psychometric instruments assessing pet bereavement and related caregiver psychological constructs: the Pet Bereavement Questionnaire (PBQ) (Hunt and Padilla, 2006)(Hunt and Padilla, 2006), which measures the psychological impact of losing a pet across grief, anger, and guilt dimensions using 4-point Likert rating scales for items (0 = strongly disagree, 3 = strongly agree); the Lexington Attachment to Pets Scale (LAPS) (Johnson et al., 1992), measuring attachment to pets on the same type of 4-point Likert scale; the Animal–Human Continuity Scale (AHCS) (Templer et al., 2006), evaluating perceptions of human–animal continuity on a 7-point rating Likert scale (1 = strongly disagree, 7 = strongly agree); the Positivity Scale (P-Scale) (Caprara et al., 2012) assessing general optimism and satisfaction on a 5-point scale (1 = strongly disagree, 5 = strongly agree); and the Testoni Death Representation Scale (TDRS) (Testoni et al., 2017), which explores conceptualizations of death along a continuum from death as passage to death as annihilation (high scores indicate annihilation) using a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree). For all scales, scoring followed the procedures described in the original MDQ validation study, with higher scores indicating stronger levels of the respective constructs being assessed (e.g., grief-related distress, attachment to pets, human–animal continuity, positivity, or death representation as annihilation).

The psychological instruments included in the MDQ were selected to capture complementary dimensions of the bereavement experience, including grief-related distress, attachment, cognitive representations of death, and broader attitudinal and affective responses. This multidimensional framework allows for a structured characterization of caregiver grief following the loss of a companion dog and provides the basis for examining how these caregiver-related dimensions may relate to the reported responses of surviving dogs. The translation procedure and preliminary measurement checks of the psychometric instruments are reported in Supplementary File 1 and Table S1 (Uccheddu et al., 2019).

Statistical analysis

Pearson correlation coefficients were used to preliminarily examine associations between all quantitative and binary study variables. Pearson's chi-square (χ^2) tests of independence in 2×2 contingency tables, with Yates' continuity correction as necessary, were employed for categorical variables; Fisher's exact test was used when expected cell counts were below 5. Multiple logistic regression was used for binary and ordinal dependent variables; model fit was evaluated using Nagelkerke R^2, and results are reported as odds ratios [Exp(B)] with 95% confidence intervals (CIs). Multiple linear regression analysis was conducted for quantitative dependent variables, with model fit assessed using R^2. Unstandardized (B) coefficients, along with 95% confidence intervals (CIs) and standardized (β), were reported.

Data analysis followed a three-step procedure: preliminary associations (Step 1), regression models (Step 2) and *post hoc* exploratory associations (Step 3).

The variables considered in Step 1 were not selected as an unrestricted or atheoretical screening set. Rather, they were chosen because they were either central to the present research question, namely euthanasia and expected death, or had previously shown relevance to caregiver grief-related responses and/or reported behavioral changes in surviving dogs within the same MDQ-based research program. Step 1 analyses were conducted as part of a conservative strategy to guide variable selection and reduce the risk of overfitting; thus, no corrections for multiple testing were applied. These preliminary analyses were used to further filter theoretically and empirically relevant variables, and only variables meeting the predefined association criterion ($p \leq 0.05$) were retained for inclusion as independent variables in the Step 2 regression models, except for euthanasia (yes/no) and expected death (yes/no), which were retained by design. Each MDQ construct served as the dependent variable. The three-step procedure was applied sequentially to caregivers and then to surviving dogs as follows. Preliminary correlations were assessed between caregiver MDQ construct scores (PBQ_Grief, PBQ_Anger, PBQ_Guilt, PBQ_Total, LAPS_GeneralAttachment, LAPS_PeopleSubstituting, LAPS_AnimalRights, LAPS_Total, AHCS_Total, P_Scale, and TDRS_Total), the two contextual death-related variables (euthanasia, expected death), and demographic factors (country, gender, age, education, marital status, household composition, presence and number of children, and garden access).

Step 1 associations were then examined between the surviving dog's behavioral and emotional changes (changes in playing, sleeping, eating, fearfulness, vocalization, attention-seeking, activity levels, avoidance of the deceased dog's resources, the most persistent change, and its duration), the two contextual death-related variables (euthanasia, expected death), and variables relating to the dog's relationship with the deceased companion (time spent together, shared activities and resources, relationship type), which had shown significant associations in our previous studies (Uccheddu et al., 2022). For each behavior/emotional state, responses were first recorded into a binary variable by identifying the most frequently reported change (either increase or decrease) and grouping all other responses (e.g., no change, don't know) as the reference category ("no change"). This approach allowed us to focus on the

predominant direction of behavioral change for each item.

MDQ constructs that in caregiver Step 2 analyses were significantly predicted either by euthanasia or expected death, as well as country of residence (Italy vs. UK) and whether the surviving dog saw the deceased's body were deemed relevant to the study aims and thus also included in this preliminary correlation matrix. To ensure model parsimony and interpretability while minimizing the risk of overfitting, except for euthanasia and expected death, which were included by default, only variables that had been previously identified as relevant (Uccheddu et al., 2022, 2019) and were significantly associated ($p \leq 0.05$) with at least one canine behavioral outcome in these preliminary correlations were retained for inclusion in Step 2 regression models, with surviving dog behavioral changes serving as dependent variable. Additionally, when both a total score (e.g., PBQ_Total, LAPS_Total) and its individual dimensions showed significant Step 1 associations with the outcome variables, only the individual dimensions were included in the regression models. This strategy allowed for a more precise identification of the specific factors driving the observed effects.

Lastly, for both the caregivers and surviving dogs, exploratory analyses were performed post hoc (Step 3) to provide further insights that could help interpret significant regression findings. Specifically, we tested the association between euthanasia and expected death, as well as associations between these two contextual death-related variables (included as outcome variables) and selected caregiver and dog characteristics. Between-country differences (Italy vs. UK) in variables that showed significant associations were then examined using independent-samples t-tests for continuous variables, and with Pearson's chi-square test of independence for categorical variables. When Levene's test indicated unequal variances, the unequal-variance estimate was used. Effect sizes were reported as Cohen's d for continuous variables (0.2 = small, 0.5 = medium, 0.8 = large) and Cramer's V for categorical variables.

All statistical analyses were conducted using IBM SPSS Statistics 29 (USA) for descriptive, correlations and regression analysis, and the R package lavaan for CFAs. All values of $p \leq 0.05$ were considered statistically significant.

Results

We collected reports from 428 Italian caregivers and 227 UK caregivers.

Descriptive statistics

Preliminary descriptive statistics for the quantitative study variables indicated good absolute values for skewedness and kurtosis (non-normality thresholds: skewedness > 2, kurtosis > 7), supporting the assumption of normal distribution.

Tables 1 and 2 report the descriptive statistics of the variables included in subsequent analyses for caregivers and their surviving dogs, respectively, in the total sample (N = 655). As for the deceased dog, about 64.0% of participants (n = 419) reported having lost their dog over one year before, 22.0% (n = 146) less than 6 months before, and 14.0% (n = 90) between 6 and 12 months before participating in the survey. Most deceased dogs were acquired (purchased or adopted) between 51 and 70 days of age (32.7%, n = 214), were purebred (54.4%, n = 356), and medium-sized (40.9%, n = 268). Females (52.4%, n = 343) and neutered dogs (71.0%, n = 465) were the most represented. Most caregivers indicated that the deceased dog was not their first (78.3%, n = 513). The dog's mean age at death was 11.40 ± 3.90 (SD) years (range 1.00–19.00; skewness = –0.94, kurtosis = –0.94) and, on average, they had lived with their caregiver for 9.70 ± 4.50 (SD) years (range 1.00–18.00; skewness = –0.25, kurtosis = –0.75). Descriptive statistics for the human and canine populations from Italy (n = 428) and the UK (n = 227) are provided separately in the Supplementary Materials (Tables S2-S5), together with between-country comparisons for the main caregiver-, deceased dog-, and surviving dog-related variables (Tables S6–S7).

Table 1
Descriptive statistics of caregiver variables: demographics and MDQ (Mourning Dog Questionnaire) constructs (N = 655).

	Score range	N (%)	Skewness	Kurtosis
Caregiver demographic characteristics				
Country (IT)		428 (65)		
Gender (Female)		593 (90)		
Marital status (Married/ cohabitant)		403 (61)		
Household composition (Alone)		134 (20)		
Presence of children (Yes)		141 (21)		
Garden (Yes)		530 (81)		
		Mean (SD)		
Age (years)	18–74	43.59 (11.48)	0.092	-0.594
Education (years)	8–20	16.16 (3.10)	-0.466	-0.854
Caregiver MDQ variables:				
PBQ_Grief	0–21	14.91 (4.56)	-0.695	-0.223
PBQ_Anger	0–15	3.26 (3.00)	1.000	0.757
PBQ_Guilt	0–12	3.84 (3.38)	0.756	-0.371
PBQ_Total	1–48	22.00 (8.45)	0.126	-0.329
LAPS_General Attachment	6–33	29.00 (3.74)	-1.385	2.625
LAPS_People Substituting	0–21	14.32 (4.73)	-0.547	-0.245
LAPS_Animal Rights	1–15	13.57 (1.98)	-2.294	6.917
LAPS_Total	7–69	56.88 (8.93)	-1.074	1.769
AHCS_Total	34–83	67.23 (7.76)	-0.807	1.053
P_Scale	10–40	28.57 (6.09)	-0.480	0.108
TDRS_Total	6–30	19.69 (6.51)	0.081	-0.843
		N (%)		
Euthanasia (vs unassisted)		436 (67)		
dog's behavioral issues		3 (1)		
dog's health issues		433 (99)		
Expected Death (yes vs no)		317 (48)		

For demographic variables, only the most frequently selected option is reported. See Supplementary material for full details. A copy of the MDQ form, including all response options, is available in (Uccheddu et al., 2019). PBQ: Pet Bereavement Questionnaire; LAPS: Lexington Attachment to Pets Scale; AHCS: Animal–Human Continuity Scale; P-Scale: Positivity Scale; TDRS: Testoni Death Representation Scale.

Caregiver grief-related responses

Only the significantly correlated variables relevant to the aims of the present study are described here (refer to Table 3 for full details). Briefly, PBQ Anger, PBQ Guilt, and PBQ Total, as well as scores on LAPS Animal Rights and the AHCS Total were negatively correlated with Euthanasia (1 = yes). All caregiver PBQ pet grief distress scores (i.e., PBQ Grief, Anger, Guilt, and Total) and, marginally, the P-Scale score was negatively correlated with an expected death (1 = yes). Higher levels of PBQ Grief were associated with being younger, female, less educated, living in the UK, and lacking access to a house garden. Similarly, PBQ Anger scores were higher among younger participants with lower education levels, no garden access, and those living in Italy. PBQ Guilt was significantly higher among younger and less educated caregivers, those living in Italy, not in a committed relationship, and without garden access. The PBQ Total score followed a similar pattern, showing negative associations with age, education, garden access, and UK residency.

Table 2
Descriptive statistics of surviving dog variables: demographics and behavioral characteristics assessed via the MDQ (Mourning Dog Questionnaire) (N = 655).

	Range	N (%)	Skewness	Kurtosis
Surviving dog demographic characteristics				
Sex (Female)		387 (59)		
Sexual status (Neutered)		440 (67)		
Age at acquisition (51–70 days old)		228 (35)		
Breed (Pure)		359 (55)		
Size (Medium)		294 (45)		
Time spent with the caregiver (> 12 months)		624 (95)		
First dog (No)		611 (93)		
		Mean (SD)		
Age of surviving dog when the companion dog died (years)	1–22	7.31 (3.86)	0.543	-0.169
Dog-dog relationship variables:				
Shared activities – Sleeping*	0–2	1.63 (0.61)	-1.434	0.918
Shared activities – Fighting*	0–2	0.47 (0.67)	1.127	0.005
Shared activities – Grooming*	0–2	0.88 (0.78)	0.215	-1.312
Shared activities – Playing*	0–2	1.40 (0.69)	-0.717	-0.657
Dog-dog relationship – Friendly**	1–5	4.17 (1.15)	-1.441	1.198
Dog-dog relationship – Agonistic**	1–5	2.22 (1.21)	0.664	-0.561
Dog-dog relationship - Mutual Tolerance**	1–5	3.20 (1.44)	-0.251	-1.304
Dog-dog relationship – Parental**	1–5	3.05 (1.45)	-0.018	-1.331
		N (%)		
Time spent together (>12 months)		606 (92%)		
Resource sharing - food (Yes)		261 (40%)		
Resource sharing - resting area (Yes)		577 (88%)		
Resource sharing - objects (Yes)		424 (65%)		
Saw body of deceased dog		300 (46%)		
Surviving dog's behavioral changes:				
Playing				
Yes (Less)		391 (60%)		
Yes (More)		34 (5%)		
No		214 (33%)		
I do not know		16 (2%)		
Sleeping				
Yes (Less)		57 (9%)		
Yes (More)		262 (40%)		
No		303 (46%)		
I do not know		33 (5%)		
Eating				
Yes (Less)		200 (30%)		
Yes (More)		30 (5%)		
No		417 (64%)		
I do not know		8 (1%)		
Fearfulness				
Yes (Less)		31 (5%)		
Yes (More)		236 (36%)		
No		363 (55%)		
I do not know		25 (4%)		
Vocalizations				
Yes (Less)		34 (5%)		
Yes (More)		198 (30%)		
No		405 (62%)		
I do not know		18 (3%)		
Elimination behaviors				
Yes (Less)		20 (3%)		
Yes (More)		35 (5%)		
No		532 (81%)		
I do not know		68 (10%)		
Attention-seeking				
Yes (Less)		56 (8%)		
Yes (More)		445 (68%)		
No		146 (22%)		
I do not know		8 (1%)		
Activity level				
Yes (Less)		299 (46%)		
Yes (More)		73 (11%)		
No		267 (41%)		
I do not know		16 (2%)		
Avoidance of deceased dog's resources (Yes)		116 (18%)		
		Mean (SD)		
Most persistent behavioral change				
Fearfulness		127 (19%)		
Eating		120 (18%)		
Sleeping		102 (16%)		
Vocalizations		85 (13%)		
Playing		75 (12%)		
Attention-seeking		59 (9%)		
Activity level		15 (2%)		
No change		72 (11%)		
Duration of the most persistent behavioral change (N = 652)				
Over 6 months		179 (27%)		
2 months		101 (15%)		
2 weeks-1 month		100 (15%)		
3–4 months		74 (11%)		
1–2 weeks		44 (7%)		
5–6 months		37 (6%)		
1–6 days		34 (5%)		
No change		83 (13%)		

Note: For demographic variables, only the most frequently selected option is reported. See Supplementary material for full details. A copy of the MDQ form, including all response options, is available in (Uccheddu et al., 2019). * Shared activities were rated on a 3-point Likert scale: 0 (never), 1 (sometimes), 2 (often). ** The dog-dog relationship was assessed using a Likert scale ranging from 1 (totally disagree) to 5 (totally agree).

With respect to attitudinal and cognitive constructs, LAPS Animal Rights scores were higher among less educated participants and those living in Italy. LAPS People Substituting scores were higher in UK participants and lower among those in a stable relationship and with higher education. LAPS Total was positively associated with UK residency and

Table 3
Pearson correlations between caregiver grief variables and demographic and death-related factors assessed via the MDQ (Mourning Dog Questionnaire) (N = 647).

Caregiver grief variables	Demographic and death-related factors							
	Country	Gender	Age (years)	Education (years)	Marital status	Garden	Expected death	Euthanasia
PBQ_Grief	0.026	0.082*	-0.093*	-0.187**	0.010	-0.096*	-0.135*	-0.062
PBQ_Anger	-0.108*	0.052	-0.129*	-0.128**	-0.029	-0.096*	-0.142*	-0.254*
PBQ_Guilt	-0.213*	-0.007	-0.184*	-0.058	-0.105*	-0.098*	-0.095*	-0.166*
PBQ_Total	-0.110*	0.060	-0.170*	-0.169**	-0.047	-0.125*	-0.161*	-0.190*
LAPS_General Attachment	0.053	0.060	0.067	-0.123**	-0.023	-0.011	-0.061	-0.014
LAPS_People Substituting	0.200*	0.063	-0.032	-0.092*	-0.081*	0.074	-0.015	0.005
LAPS_Animal Rights	-0.108*	-0.003	0.044	-0.141**	-0.041	-0.058	-0.051	-0.117*
LAPS_Total	0.106*	0.058	0.020	-0.132**	-0.062	0.023	-0.045	-0.029
AHCS_Total	-0.074	0.048	0.008	-0.091*	-0.053	-0.142*	-0.024	-0.120*
P_Scale	-0.037	-0.029	0.052	0.097*	0.166*	-0.024	-0.084*	-0.044
TDRS_Total	0.057	-0.051	0.030	0.017	-0.022	0.068	0.051	-0.023

Note. Codes for binary variables: Country (0 = Italy; 1 = UK); Gender (0 = Male; 1 = Female); Marital status (0 = Not married/cohabiting; 1 = Married/cohabiting); Garden (0 = No; 1=Yes); Death expected (0 = No; 1 = Yes); Euthanasia (0 = No, 1 = Yes). * Correlation is significant at the 0.05 level (2-tailed), ** Correlation is significant at the 0.01 level (2-tailed). PBQ = Pet Bereavement Questionnaire. LAPS = Lexington Attachment to Pets Scale. AHCS = Animal Human Continuity Scale. P-scale= positivity scale.

negatively associated with education. AHCS Total scores were significantly higher in less educated participants and those without garden access. Finally, P-Scale scores were higher among those in a committed relationship, while they were positively associated with education level.

Multiple linear regression model results are summarized in Table 4 in which only statistically significant factors are reported. Euthanasia and expected death significantly predicted lower caregiver grief distress. Specifically, euthanasia was associated with reduced PBQ Anger and PBQ Guilt, while an expected death predicted lower PBQ Grief, Anger, and Guilt. These death-related circumstances were also associated with constructs relating to other caregiver psychological constructs Euthanasia predicted lower endorsement of animal rights and weaker perceptions of human–animal continuity, while expected death predicted lower caregiver positivity.

Several demographic variables also emerged as significant predictors of MDQ constructs. Country of residence played a role, with UK participants reporting higher Grief but lower Guilt than Italian participants. Female gender was associated with higher PBQ Grief, while being married or in a committed relationship predicted lower PBQ Guilt and higher P-Scale scores (B = 1.974, p = 0.001). Younger age significantly predicted higher scores across all PBQ subscales. Lower education predicted higher PBQ Grief, stronger animal rights concern, and lower positivity. Lastly, lack of garden access was associated with increased PBQ Grief and stronger perceptions of human–animal continuity.

With respect to the post hoc analyses carried out to improve regression result interpretation, a significant association between euthanasia and expected death was found ($\chi^2 = 35.25$, df = 1, p = 0.001) with 77.6% (n = 338) of euthanized dogs having had a death that was expected by the caregiver, compared to only 22.4% (n = 49) of dogs who died without euthanasia. The dog's euthanasia and expected death were

Table 4
Multiple Linear Regression results for Caregiver grief variables by demographic and death-related factors assessed via the MDQ (Mourning Dog Questionnaire) (N=647).

Caregiver grief variables	Predictors	B	95% CI	Beta	p-value	R-square	F (8, 638)	p-value
PBQ_Grief						0.085	7.414	0.001
	Expected death	-0.996	[-1.696; −0.297]	-0.110	0.005			
	Country	1.261	[0.466; 2.055]	0.133	0.002			
	Gender	1.330	[0.141; 2.520]	0.083	0.028			
	Age (years)	-0.045	[-0.075; −0.015]	-0.115	0.003			
	Education (years)	-0.294	[-0.409; −0.179]	-0.201	0.001			
	Garden	-1.128	[-2.034; −0.222]	-0.098	0.015			
PBQ_Anger						0.099	8.793	0.001
	Expected death	-0.510	[0.967; −0.053]	-0.085	0.029			
	Euthanasia	-1.329	[-1.827; −0.830]	-0.210	0.001			
	Age (years)	-0.028	[-0.048; −0.009]	-0.109	0.005			
	Education (years)	-0.087	[-0.162; −0.011]	-0.090	0.024			
PBQ_Guilt						0.092	8.109	0.001
	Death expected	-0.518	[-1.035; −0.001]	-0.077	0.049			
	Euthanasia	-0.636	[-1.199; −0.072]	-0.089	0.027			
	Country	-1.114	[-1.701; −0.527]	-0.158	0.001			
	Age (years)	-0.041	[-0.063; −0.019]	-0.140	0.001			
	Marital status	-0.581	[-1.098; −0.065]	-0.084	0.027			
LAPS_Animal Rights						0.038	3.124	0.002
	Euthanasia	-0.351	[-0.683; −0.019]	-0.086	0.038			
	Education (years)	-0.064	[-0.114; −0.014]	-0.103	0.012			
AHCS_Total						0.041	3.411	0.001
	Euthanasia	-1.729	[-3.054; −0.405]	-0.106	0.011			
	Garden	-2.516	[-4.090; −0.943]	-0.129	0.002			
P_Scale						0.043	4.625	0.001
	Expected death	-1.061	[-2.019; −0.103]	-0.087	0.030			
	Education (years)	0.255	[0.098; 0.412]	0.129	0.002			
	Marital status	1.974	[1.017; 2.931]	0.157	0.001			

Note. Codes for binary variables: Country (0=Italy; 1=UK); Gender (0=Male; 1=Female); Marital status (0=Not married/cohabiting; 1=Married/cohabiting); Garden (0=No; 1=Yes); Death expected (0=No; 1=Yes); Euthanasia (0=No, 1=Yes). PBQ = Pet Bereavement Questionnaire. LAPS = Lexington Attachment to Pets Scale. AHCS = Animal Human Continuity Scale. P-scale= positivity scale.

also significantly associated with some caregiver variables. Specifically, euthanasia was positively correlated with caregiver age ($r = 0.094$, $p = 0.017$) and a higher education level ($r = 0.120$, $p = 0.002$). The death being expected was also positively associated with education ($r = 0.108$, $p = 0.006$). Both expected death ($r = 0.216$, $p = 0.001$) and euthanasia ($r = 0.138$, $p = 0.001$) were positively correlated with the amount of time the caregiver had spent with the deceased dog.

Surviving dog reported behavioral and emotional changes

Results are summarized in Table 5, where only variables showing significant correlations are reported. Euthanasia and expected death of the deceased dog were not directly correlated with any specific behavioral or emotional change in the surviving dog. However, several behavioral outcomes were significantly correlated with caregiver grief dimensions (PBQ subscales), as well as perceptions (AHCS) and attitudes (LAPS Animal Rights), which had themselves been significantly predicted by these death-related variables in previous regression models.

Reports of dogs sleeping more, increased fearfulness, and avoidance of the deceased dog's resources were positively correlated with both caregivers' PBQ Grief and PBQ Anger scores. Playing less, eating less, increased vocalizations, attention-seeking, and reduced activity levels in the surviving dogs were positively correlated with higher caregiver PBQ Grief scores, while a longer duration of the most persistent behavioral change was positively correlated with PBQ Anger scores. Eating less was significantly correlated with higher scores on both the LAPS Animal Rights and AHCS Total scales and negatively correlated with TDRS Total scores. Sleeping more, increased attention-seeking, and avoidance of the deceased dog's resources were positively correlated with LAPS Animal Rights scores. Conversely, playing less and reduced activity levels were negatively correlated with TDRS Total scores.

Country of residence was also associated with behavioral outcomes: being from the UK was positively associated with increased sleeping and a longer duration of the most persistent behavioral change. Finally, several significant correlations were found with variables related to the dogs' relationship prior to the loss. Playing less was positively associated with longer cohabitation (>12 months), all three types of shared activities (sleeping, grooming, playing), friendly and parental relationships, and all types of resource-sharing (food, resting area, and objects); while it was negatively associated with a mutually tolerant relationship. Sleeping more was positively correlated with longer cohabitation, grooming and playing together, a friendly relationship, and all types of resource sharing (food, resting area, and objects). Eating less was associated with grooming and playing together, a parental relationship, and all resource-sharing variables (food, resting area). Fearfulness was positively associated with a parental relationship (i.e. surviving dogs were the offspring of the deceased). An increase in attention-seeking was positively correlated with playing together, a friendly and parental relationship, and certain types of resource sharing (resting area and objects). Reduced activity levels were positively correlated with longer cohabitation, all shared activities (sleeping, grooming, playing), a friendly and parental relationship, and all types of resource sharing (food, resting area, and objects). Avoidance of the deceased dog's resources was negatively associated with sharing food and objects, as well as playing. A longer duration of the most persistent behavioral change was positively associated with longer cohabitation, shared sleeping, grooming, playing, friendly and parental relationship, and sharing resting area and objects.

Full multiple logistic regression and linear regression model results, including effect sizes and confidence intervals for all significant predictors of the surviving dog's responses, are presented in Table 6, which reports only significant associations. Data are shown only when at least one of the significant predictors included a death-related contextual variable or a caregiver grief dimension (PBQ) that had itself been

Table 5
Pearson correlations between the surviving dog's behavioral changes and country, circumstances of dog's death, caregiver emotional states and dog-dog relationship factors assessed via the MDQ (Mourning Dog Questionnaire) (N=652).

	Surviving dog's behavioral changes:							Avoid Deceased Dog's Resources	Duration of Behavioral Change
	Playing Less	Sleeping More	Eating Less	Fearful Level More	Vocalization More	Attention Seeking More	Level Activity Less		
Caregiver MDQ constructs predicted by euthanasia and/or expected death									
PBQ_Grief	0.125*	0.128*	0.148*	0.153*	0.093*	0.157*	0.089*	0.086*	0.047
PBQ_Anger	0.028	0.102*	0.033	0.132*	-0.003	0.055	0.049	0.123*	0.085*
LAPS_Animal Rights	0.013	0.094*	0.106*	0.036	0.003	0.111*	0.040	0.120*	0.021
AHCS_Total	0.025	0.039	0.106*	0.021	-0.009	0.031	0.057	0.069	0.061
TDRS_Total	-0.098*	-0.092	-0.108*	-0.036	-0.044	-0.043	-0.109*	-0.010	0.022
Country	0.074	0.152*	-0.053	0.041	0.017	0.026	-0.005	-0.054	0.104*
Time dogs spent together	0.146*	0.114*	0.038	-0.028	0.074	0.066	0.156*	0.041	0.114*
Shared activities between dogs									
Sleeping	0.194*	0.084	0.070	0.004	0.032	0.047	0.111*	-0.073	0.116*
Grooming	0.202*	0.157*	0.131*	0.020	0.036	0.018	0.104*	-0.027	0.121*
Playing	0.380*	0.190*	0.108*	0.057	0.051	0.087*	0.227*	-0.098*	0.192*
Type of dog-dog relationship									
Friendly	0.189*	0.108*	0.053	0.009	0.003	0.115*	0.110*	-0.065	0.136*
Mutual Tolerance	-0.112*	-0.003	0.054	0.009	0.011	0.031	0.000	0.037	0.019
Parental	0.156*	0.075	0.164*	0.126*	0.063	0.087*	0.127*	0.017	0.151*
Resource sharing - food	0.108*	0.144*	0.163*	-0.014	-0.040	-0.022	0.105*	-0.085*	0.027
Resource sharing - resting area	0.170*	0.118*	0.102*	0.050	0.057	0.081*	0.139*	-0.014	0.096*
Resource sharing - objects	0.298*	0.153*	0.069	0.041	0.034	0.096*	0.150*	-0.113*	0.180*

Note. Codes for binary variables: Country (0 = Italy; 1=UK); Death expected (0 = No; 1 = Yes); Euthanasia (0 = No, 1 = Yes); Time dogs spent together (0 = Less than 12 months; 1 = Over 12 months); Resource sharing (0 = No; 1 = Yes); Surviving dog's behavioral changes (0 = No; 1 = Yes); Avoid Deceased Dog's Resources (0 = No; 1 = Yes). * Correlation is significant at the 0.05 level (2-tailed), ** Correlation is significant at the 0.01 level (2-tailed). PBQ = Pet Bereavement Questionnaire. LAPS = Lexington Attachment to Pets Scale. AHCS = Animal Human Continuity Scale. TDRS= Testoni Death Representation Scale. P-scale= positivity scale.

Table 6
Multiple logistic regression models predicting dog behavior's changes assessed via the MDQ (Mourning Dog Questionnaire).

Dependent variables	Predictors	B	Exp(B)	95% CI	p-value	R-square
Sleeping more						0.149
	Country	0.634	1.885	[1.317; 2.699]	0.001	
	PBQ_Anger	0.080	1.083	[1.023; 1.146]	0.006	
	Time dogs spent together	0.996	2.708	[1.281; 5.727]	0.009	
	Shared Playing (often)	0.928	2.529	[1.284; 4.982]	0.007	
Eating less						0.112
	PBQ_Grief	0.051	1.053	[1.010; 1.097]	0.016	
	Dog-dog Parental relationship (agree)	0.785	2.193	[1.189; 4.045]	0.012	
	Dog-dog Parental relationship (strongly agree)	0.903	2.466	[1.415; 4.297]	0.001	
	Resource sharing (food)	0.562	1.754	[1.234; 2.494]	0.002	
Increased fearful level						0.054
	PBQ_Grief	0.071	1.073	[1.021; 1.128]	0.005	
	Dog-dog parental relationship (strongly agree)	0.576	1.778	[1.089; 2.904]	0.021	
More vocalization						0.013
	PBQ_Grief	0.047	1.048	[1.009; 1.089]	0.016	
More attention seeking						0.058
	PBQ_Grief	0.066	1.068	[1.030; 1.108]	0.001	
Avoid the deceased dog's resources						0.078
	PBQ_Anger	0.091	1.096	[1.026; 1.170]	0.006	
	LAPS_Animal Rights	0.199	1.220	[1.054; 1.411]	0.008	
	Resource sharing (objects)	-0.499	0.607	[0.393; 0.937]	0.024	
Duration Behavioral Change						0.073
	PBQ_Anger	0.052	1.049	1.002; 1.099]	0.015	
	Country (UK vs IT)	0.477	1.611	[1.163; 2.231]	0.004	
	Time dogs spent together (over 12 months vs less than 12 months)	0.654	1.924	[01.137; 3.254]	0.015	
	Shared Playing (often vs never)	0.553	1.739	[1.008; 2.997]	0.047	
	Dog-dog parental relationship (strongly agree vs strongly disagree)	0.586	1.797	[1105; 2.921]	0.018	

Note. Codes for binary variables: Country (0 = Italy; 1 = UK); Death expected (0 = No; 1 = Yes); Euthanasia (0 = No, 1 = Yes); Time spent together dogs (0 = Less than 12 months; 1 = Over 12 months); Resource sharing (0 = No; 1 = Yes); Surviving dog's behavioral changes (0 = No; 1 = Yes); Avoid Deceased Dog's Resources (0 = No; 1 = Yes). PBQ = Pet Bereavement Questionnaire. LAPS = Lexington Attachment to Pets Scale.

previously predicted by these factors. Neither euthanasia nor expected death significantly predicted any of the reported behavioral changes in the surviving dog. However, caregiver pet grief emerged as a key predictor of several changes. Specifically, higher PBQ Grief predicted an increased likelihood of eating less, displaying greater fearfulness, vocalizing more, and seeking more attention. Additionally, PBQ Anger predicted increased sleeping, avoidance of the deceased dog's resources, and longer duration of behavioral changes.

Several contextual and relational confounds emerged for canine behaviors. Sleeping more was more likely in dogs from the UK, in dogs that had cohabited longer with the deceased dog, and in those that had frequently played with the deceased dog. Eating less and greater fearfulness were more likely in surviving dogs described as having had a parent-offspring relationship with the deceased and in those who had shared food before the loss.

Avoidance of the deceased dog's resources was more likely in dogs whose caregivers scored higher on LAPS Animal Rights. In contrast, dogs that had shared objects with the deceased dog before the loss were less likely to show avoidance behavior. In contrast, dogs that had shared objects with the deceased dog before the loss were significantly less likely to show avoidance behavior.

Longer-lasting behavioral changes were more likely in dogs that had cohabited with the deceased dog for more than 12 months, had frequently played with the deceased dog, or had a strong parent–offspring bond with the deceased dog. Being from the UK, as opposed to Italy, was also associated with longer-lasting behavioral changes.

A positive association was found between a dog's age at death and both euthanasia (mean age 11.99 years old ± 3.50 SD, $r = 0.190$, $p = 0.001$) and expected death (mean age 12.63 years old ± 3.35 SD, $r = 0.297$, $p = 0.001$). Seeing the body of the deceased dog was significantly less common when death occurred through euthanasia (35.8% vs 65.3% in non-euthanasia cases; $\chi^2 = 51.51$, df = 1, $p < 0.001$). Moreover, neutered dogs were more frequently euthanized (70.5%) compared to intact dogs (55.3%) ($\chi^2 = 14.05$, $p = 0.001$). Euthanasia was also more common among larger dogs ($r = 0.112$, $p = 0.004$) and those acquired at an older age ($r = 0.120$, $p = 0.002$). Lastly, whether the deceased dog was the caregiver's first dog was significantly associated with both euthanasia ($\chi^2 = 8.01$, $p = 0.005$) and expected death ($\chi^2 = 7.34$, $p = 0.007$): first dogs were more often euthanized (76.1%) than non-first dogs (63.4%) and their deaths were more frequently reported as expected (58.5%) compared to dogs whose caregivers had previously lived with other dogs (45.6%).

Discussion

In this study we found that the two contextual variables related to a pet dog's death, namely euthanasia and whether the death was expected, were significantly associated with lower levels of grief-related outcomes in caregivers, as assessed by the PBQ subscales within the Mourning Dog Questionnaire (see Table 5 for a summary). These effects emerged in the presence of several demographic confounders, including the country of residence, which influenced the magnitude and direction of specific grief dimensions. Beyond grief, euthanasia and expected death were also linked to other constructs measured by the MDQ, including attitudes toward animals (LAPS Animal Rights), perceptions of human–animal continuity (AHCS), and positivity (P-Scale), which, as will be discussed below, may help to explain the grief-related results. At the same time, the two death-related variables were indirectly connected with the behavioral and emotional responses of surviving dogs: while not directly predicting such changes, they significantly covaried with caregiver grief, attitudes, and perceptions, which in turn were associated with canine outcomes.

In our study (as summarized in Tables 7 and 8), expected death was a consistent negative predictor of all PBQ dimensions, with the strongest association for Grief, where it was the only significant death-related predictor. For Anger and Guilt, the coefficients for expected death were smaller, and euthanasia also emerged as an independent predictor. These findings are broadly consistent with previous studies showing that anticipatory awareness and active participation in end-of-life decisions can mitigate emotional distress (Adams et al., 2000, 1999; Garand et al.,

Table 7
Summary of results focusing on euthanasia and expected death predicting caregiver MDQ (Mourning Dog Questionnaire) constructs.

Predictor: Euthanasia		
Outcome (direction)	Confounders (direction)	Interpretation
PBQ Anger (negative)	Age (negative), education (negative)	Euthanasia was associated with less Anger. Younger and less educated caregivers, however, were more likely to report Anger.
PBQ Guilt (negative)	Country (negative, UK < IT), age (negative), marital status (negative)	Euthanasia predicted lower guilt. The same pattern was observed for older and married/committed caregivers, while UK caregivers reported less guilt than Italians.
LAPS Animal Rights (negative)	Education (negative)	Caregivers who euthanised their dog were less likely to endorse strong animal rights views; the same effect was seen in less educated caregivers.
AHCS Total (negative)	Garden (negative)	After euthanasia, caregivers were less likely to perceive strong human–animal continuity. Lack of garden access was also associated with higher continuity perceptions.
Predictor: Expected death		
Outcome	Confounders (direction)	Interpretation
PBQ Grief (negative)	Country (positive, UK > IT), gender (positive), age (negative), education (negative), garden (negative)	Expected death was associated with lower grief. In contrast, UK caregivers, women, younger and less educated individuals, and those without garden access were more likely to report stronger grief.
PBQ Anger (negative)	Age (negative), education (negative)	Expected death predicted less Anger. Younger and less educated caregivers, however, tended to report more Anger.
PBQ Guilt (negative)	Country (negative, UK < IT), age (negative), marital status (negative)	Expected death was linked to lower guilt. Similarly, older and married caregivers reported less guilt, while UK participants showed lower guilt than Italians.
P-Scale (Positivity) (negative)	Education (positive), marital status (positive)	Expected death predicted reduced positivity. Conversely, higher education and being married/committed were associated with greater positivity.

PBQ = Pet Bereavement Questionnaire. LAPS = Lexington Attachment to Pets Scale. AHCS = Animal Human Continuity Scale. P-scale= positivity scale.

Table 8
Summary of results focusing on factors predicting surviving dog behavioral changes using the MDQ (Mourning Dog Questionnaire).

Predictor: PBQ Anger		
Outcome (direction)	Confounders (direction)	Interpretation
Sleeping more (positive)	Country (positive, UK > IT); time dogs spent together (positive); shared playing (positive)	Higher PBQ Anger predicted more sleeping in the surviving dog. Sleeping more was also more likely among UK caregivers, and in canine dyads with longer cohabitation and frequent shared play
Avoiding deceased dog's resources (positive)	LAPS Animal Rights (positive); resource sharing – objects (negative)	Higher PBQ Anger predicted avoidance of the deceased dog's resources. This behavior was also more likely among dogs whose caregivers had stronger animal rights views, and less common when the two dogs had previously shared objects.
Duration of behavioral change (positive)	Country (positive, UK > IT); time dogs spent together > 12 months (positive); shared playing – often (positive); dog–dog parental relationship – strongly agree (positive)	Higher PBQ Anger predicted longer persistence of behavioral changes. Duration of changes was also positively associated with being from the UK, longer cohabitation between the two dogs, frequent play, and parental-type relationships.
Predictor: PBQ Grief		
Outcome (direction)	Confounders (direction)	Interpretation
Eating less (positive)	Dog–dog parental relationship (positive); resource sharing – food (positive)	Higher PBQ Grief predicted reduced food intake. This behavioral change was also more likely when there was a parental relationship with the deceased dog and when the two dogs used to share food.
Increased fearfulness (positive)	Dog–dog parental relationship (positive)	Higher PBQ Grief predicted greater fear in the surviving dog, and this was also more likely when there was a parental relationship with the deceased dog.
More vocalization (positive)	—	Higher PBQ Grief predicted more vocalization in the surviving dog.
More attention seeking (positive)	—	Higher PBQ Grief predicted greater attention-seeking behaviors.

2012; McCutcheon and Fleming, 2002). At the same time, they highlight different nuances: while expecting the death appears to attenuate caregiver grief, likely because it allows time to prepare for the loss, euthanasia carries an additional and specific effect, particularly in reducing anger and, to a lesser extent, guilt.

Importantly, in our sample euthanasia and expected death were often intertwined, with a significant association between the two variables. Indeed, 77.6% of euthanized dogs had a death that was expected by the caregiver, compared to only 22.4% of dogs who died without euthanasia. This overlap between euthanasia and expected death suggests that being both prepared for the death and actively involved in the end-of-life decision-making process, by far the most common condition in our study, may buffer negative emotional responses, specifically those associated with distress and self-blame (anger and guilt), fostering a greater sense of control, emotional preparedness, and closure. In this light, euthanasia may be experienced not only as a medical act but also as an expression of compassionate agency. In our sample, euthanasia was the leading cause of death, consistent with previous reports (Pegram et al., 2021; Previti et al., 2024), and was positively associated with both the dog's age and whether the death was expected. In other words, dogs whose death was expected and medically assisted were typically senior individuals, irrespective of country of residence. This is not surprising, as old age is recognized as a major risk factor for euthanasia (Gorodetsky, 1997; Michell, 1999), which likely goes along with the greater emphasis on animal well-being in modern societies, where euthanasia is frequently conceptualized as a compassionate act aimed at preventing unnecessary suffering (Powell et al., 2021), particularly in the geriatric phase of life (Ukraintseva et al., 2016; McKenzie and Chen, 2022; Pirrone et al., 2025). Indeed, as Cooney and Kipperman (2023) and Stokes et al. (2002) argue, if life is to be ended, it should be ended at the appropriate time and for the right reasons. The higher proportion of euthanized dogs compared to those who died unassisted in our study may thus reflect a growing awareness among pet caregivers of their

animals as sentient beings, and a corresponding responsibility to protect them from avoidable suffering.

An interesting pattern was found for expected death, which, as previously noted, predicted lower grief but was also associated with poorer psychological well-being as measured by the P-Scale, a somewhat counterintuitive finding. One possible explanation is that anticipating the loss may attenuate acute grief reactions but at the same time contribute to a more prolonged or cumulative emotional strain, particularly among caregivers who witnessed a slow decline, in line with models of anticipatory grief and caregiver burden described in human caregiving (Cordaro, 2012; Garand et al., 2012; Schulz and Sherwood, 2008), where prolonged awareness of an impending death can mitigate immediate grief while potentially diminishing overall positivity and well-being. In our sample, lower psychological well-being was also predicted by not being married or in a committed relationship and by having fewer years of education, suggesting that both relational and cognitive resources might serve as protective factors. Caregivers with lower education often have lower paying jobs so higher economic burden is associated with lower wellbeing. Interestingly, these same socio-demographic variables were also associated with grief-related outcomes on the PBQ (i.e., less education predicted higher grief and anger, while marital status predicted lower guilt), pointing to a convergence whereby limited resources may not only exacerbate grief but also undermine broader psychological well-being following pet loss.

In sum, while expected death was generally associated with attenuated grief-related emotions, its association with poorer overall psychological well-being underscores the potential dual nature of anticipatory processes: they may prepare caregivers for the loss but also challenge their long-term emotional resilience, particularly in the absence of resources such as higher education or strong relational support. This finding also highlights the need to distinguish between grief-specific responses and broader mental health during the caregiving period, as sustained exposure to suffering may deplete psychological resources (Schulz and Sherwood, 2008).

Of course, interpreting these associations requires caution. Due to the correlational nature of the models, it remains unclear whether euthanasia and death expectation directly attenuate pet grief-related distress, or whether caregivers with pre-existing traits or beliefs (Packman et al., 2011), such as greater acceptance of loss or a more pragmatic view of animal death, were simply more likely to anticipate the loss or choose euthanasia. In this view, lower PBQ scores may reflect pre-existing emotional regulation, resilience, or attitudes rather than being a direct consequence of these circumstances. For example, caregivers who opted for euthanasia also reported lower concern for animal rights and weaker perceptions of human–animal continuity, suggesting a more instrumental or detached orientation. Likewise, higher education, while not directly predictive of euthanasia in this study, was associated with lower concern for animal rights and weaker perceptions of human–animal continuity, possibly reflecting more rational or cognitively oriented views of animal life and death. Thus, euthanasia may be part of a broader worldview rather than acting as a direct cause of emotional relief. Similarly, caregivers who were more emotionally resilient or more accepting of death may have been both more likely to foresee the loss and better able to cope with it, making expectation a marker rather than a cause of reduced grief (Packman et al., 2011).

As expected, country of residence was significantly associated with pet grief distress-related outcomes. UK caregivers reported lower guilt than Italians, an effect partly explained by demographic and death-related variables. Specifically, higher age, higher education, and choosing euthanasia were all associated with lower PBQ scores. Therefore unsurprisingly, caregivers from the UK were significantly older, reported more years of education, and relied more frequently on euthanasia (82.8%) than Italian caregivers. These two proportions are both consistent with those reported in the literature for the respective countries, strengthening the representativeness of our sample (Previti et al., 2024). These findings may also reflect different national framings of euthanasia. Legally, in both the UK and Italy euthanasia is restricted to veterinarians (Clutton, 2025; Cooney and Kipperman, 2023), but the cultural and regulatory contexts diverge. In Italy, animal euthanasia is permitted only when medically justified (e.g., incurable disease or dangerous behavior), otherwise it may fall under the Criminal Code (Law 189/2004), which prohibits the killing of animals "without necessity" and can lead to criminal prosecution (Gazzetta Ufficiale n, 2004). Requests based solely on caregiver preference may thus raise ethical or legal concerns, basically echoing broader cultural and moral reservations toward euthanasia in humans, that may extend to animals as well (Arathoon and Van Patter, 2024). These more conservative views on life and death may predispose caregivers toward a more conflicted stance. In the UK, by contrast, euthanasia is more common, including for older animals with chronic but non-terminal conditions (Andersen et al., 2021), and is typically conceived as part of compassionate care centered on welfare and quality of life. This conceptualization of a "good death" could be less conflictual than in the Italian context, generating less guilt in caregivers. The normalization of euthanasia in the UK, also reflected in reports showing euthanasia in nearly 90% of canine deaths, may contribute to a more pragmatic and medicalized view of end-of-life care. In our study, this cultural framing was reflected in attitudinal measures: caregivers who opted for euthanasia also reported lower concern for animal rights and weaker perceptions of human–animal continuity.

Together, these findings suggest that lower guilt in the UK may not only stem from demographics and the greater prevalence of euthanasia, but also from cultural narratives that legitimize euthanasia as a compassionate choice and frame the human–animal bond in more pragmatic terms. Differences in the social acceptability and normalization of pet-related grief might also play a role. In the UK, where companion animals are more widely considered family members and pet loss is more publicly acknowledged (Belshaw et al., 2020), caregivers may feel more comfortable expressing sadness and distress, while they may be less prone to internalize guilt. Conversely, in Italy, where pet loss might be more privately managed and less socially validated, caregivers may experience more guilt because of unresolved emotional tension or perceived personal responsibility for the death (Packman et al., 2011). Secondly, UK caregivers also reported higher psychological suffering (PBQ Grief) than those from Italy. This finding does not appear to be explained by euthanasia, which showed no association with grief, nor by death expectation, which was associated in the opposite direction and did not differ in prevalence between countries. Based on our findings, UK participants were more likely to have been living alone at the time of their dog's death, a condition previously associated with greater grief intensity in studies using the Pet Bereavement Questionnaire (Hunt and Padilla, 2006). Living alone may exacerbate grief through reduced social support, consistent with the notion of disenfranchised grief (Doka, 1999), whereby the absence of socially recognized mourning limits the availability of support.

Other individual factors, beyond those already discussed, were also found to predict pet grief, either negatively or positively. Garden access was associated with lower grief, which may be explained by the restorative effects of nature (cf. Kaplan & Kaplan's "restorative environments" theory) (Kaplan and Kaplan, 1989) or by a less emotionally intense caregiving style, in which outdoor dogs may foster more independent relationships (Pirrone et al., 2015a, 2015b). Consistently, lack of garden access was also associated with higher continuity perceptions (AHCS Total). Female caregivers reported greater pet grief, in line with previous findings (Uccheddu et al., 2019), which may reflect women's general tendency to experience stronger grief responses than men (McCutcheon and Fleming, 2002; Stokes et al., 2002), and their greater likelihood of seeking support (Behler et al., 2020; Turner, 1997). Nevertheless, it remains unclear whether bereaved female caregivers truly experience more intense grief than their male counterparts, or whether they are simply more ready and willing to report it (Stroebe et al., 2001)

These findings have clear clinical implications. Veterinarians should

recognize not only the grief of caregivers but also the possible behavioral and emotional changes in surviving dogs. Considering both aspects can improve how they communicate and support caregivers during this time. Cultural differences are also important. As shown in the contrast between Italy and the UK, national rules and cultural views on euthanasia can strongly influence how grief and guilt are experienced. Veterinary communication and bereavement support should therefore be adapted to cultural contexts, so that caregivers feel understood and supported in their emotional responses.

The findings that the PBQ measures of Grief, Anger and Guilt are negatively related to the caregiver's age, especially in the situation where the dog's death was expected, seems to warrant some comment. However, generally, people in old age have higher happiness, less depression, better emotion regulation etc. than those in midlife (Kunzmann et al., 2014; Piazza et al., 2015; Blanchflower et al., 2023). In this study we did not take any measures of the health status of the caregivers. However, it is reasonable to assume that older individuals are more likely to have health problems and physical limitations. In the case where the dog's death was expected, it is also likely that the dog's health was also deteriorating and the dog was requiring additional attention and care, such as the administration of medications and so forth. For older caregivers, especially those with health challenges, tending to a declining pet may involve additional physical, emotional, and practical demands. The cessation of these duties might therefore contribute to a degree of caregiver relief alongside grief. However, because our study did not include direct measures of caregiver health status, caregiving burden, or feelings of relief, these factors remain plausible but untested contributors to the bereavement experience.

Euthanasia and expected death of the deceased dog were not directly correlated with any specific behavioral or emotional change in the surviving dog. This finding was highly predictable, as, from the dog's perspective, the specific circumstances of the cohabitant's death may have been less accessible or directly meaningful than the social and emotional consequences that followed. Moreover, a substantial proportion of surviving dogs (46%) had seen the body of the deceased dog, a circumstance particularly common in unassisted death cases (65.3% vs 35.8% in euthanasia cases). However, behavioral outcomes did not differ according to this exposure, providing little support for the hypothesis that dogs in this study comprehended the death event. While it remains possible that dogs possess some degree of cognitive or emotional sensitivity to death (Boyd et al., 1993), in this context, they were unlikely to be able to comprehend what had occurred, whether due to how the death was managed, the absence of direct exposure, or other environmental factors. These findings are consistent with existing literature, suggesting that the behavioral responses of surviving dogs are more strongly associated with the emotional state of the caregiver—including grief reactions shaped by factors such as euthanasia and death expectation — as well as with the loss of social interactions with the deceased dog and disruptions in daily routines, rather than with the specific circumstances or nature of the death itself (Uccheddu et al., 2022). However, while expected death and euthanasia did not directly predict behavioral changes in the surviving dog, several of these behavioral outcomes were significantly associated with caregiver grief dimensions (PBQ subscales) and attitudes (LAPS Animal Rights), which had themselves been significantly predicted by the two death-related variables in regression models (see Table 6 for a summary). This pattern suggests an indirect pathway linking death circumstances to changes in the surviving dog, possibly through the caregiver's emotional state.

Two PBQ dimensions emerged as modest but significant predictors. Higher PBQ Anger was associated with more sleeping in the surviving dog, avoidance of the deceased dog's resources, and longer persistence of behavioral changes. Higher PBQ Grief predicted reduced food intake, greater fearfulness, more vocalization, and increased attention-seeking behaviors. In other words, even though the dog may not understand or react to the nature of death *per se*, they are likely to respond to the emotional distress of the caregiver and/or its potential consequences, such as changes in caregiving behavior and daily management. Mechanisms such as human-animal emotional contagion, disruption of routine, or changes in emotional availability of the caregiver may help explain these effects (Uccheddu et al., 2022). Similar associations have previously been interpreted as evidence of social transmission (Debiec and Olsson, 2017) of emotional contagion across species, considered by some authors as a primitive form of cross-species empathy and an adaptive feature of animals living in close social groups (Yong and Ruffman, 2014). Another, not mutually exclusive, explanation is that dogs experience negative emotions themselves and seek comfort or relief from distress, as often happens in social animals exposed to shared stressors (Debiec and Olsson, 2017). Given their attention to human communicative gestures and emotional expressions, dogs may adjust their own behavior accordingly. At the same time, we cannot exclude the role of caregiver perception: caregivers experiencing stronger grief may have been more likely to interpret their dog's behavior as reflecting their own distress (Uccheddu et al., 2022). Consistently, some behavioral changes, such as sleeping more and longer persistence of alterations, were more likely reported by UK participants, who also showed higher levels of grief.

This central emotional pathway is further reinforced by additional factors that reflect the caregiver's emotional bond and cognitive framing of the loss, as caregiver attachment to the deceased dog (LAPS) also amplified reported changes, potentially through both increased sensitivity to the surviving dog's behaviors and more pronounced emotional impacts. As would be expected considering our previous study (Uccheddu et al., 2022), relational factors reflecting the depth of social bonds between the two dogs, as well as the environmental stability likely provided by their relationship, emerged as significant predictors of the surviving dog's behavioral changes as observed by the caregiver after the deadly event. Specifically, the more time the dogs had spent together and the more they had shared resources (i.e., food, resting areas, toys) and engaged in joint activities such as play, and particularly when they had an affiliative relationship such as a parent–offspring bond, the more likely the surviving dog was to show behavioral changes (i.e., reduced play and activity, increased sleep and vocalization, heightened fearfulness). This supports the idea that the loss of companionship and the disruption of shared routines play a major role in post-loss adjustment, as previously suggested (Uccheddu et al., 2019). Similar patterns have also been reported in other companion animals, including cats experiencing the loss of a cat or dog companion, suggesting that these post-loss responses are not unique to dogs or other group-living animals (Walker et al., 2016; Greene and Vonk, 2024). Such shared experiences likely function as social and environmental anchors in the dog's daily lives, and their sudden removal may lead to behavioral withdrawal or emotional dysregulation. The loss of a bonded conspecific may be particularly impactful when dogs share strong emotional attachments, reciprocal care, or closely interwoven routines, which, when disrupted, may elicit behavioral expressions akin to grief. To some extent, these results may be comparable with what has been described in human bereavement, with the intensity of post-loss reactions being predicted by relational closeness and shared daily life with the deceased person (Stroebe and Schut, 2001; Stroebe et al., 2007).

Some methodological limitations of this study should also be noted. First, the cross-sectional design, which precludes any firm conclusions about causality. It remains unclear, for instance, whether euthanasia and death expectation actively reduced caregiver grief, or whether caregivers with lower grief tendencies were more likely to anticipate death or to opt for euthanasia. Our work nonetheless opens up the potential to investigate these hypotheses. Second, as the survey was conducted and distributed online, the sample may have been subject to self-selection bias, with caregivers who had particularly intense or meaningful experiences of pet loss been more inclined to participate. This potential bias, together with the potential reporting bias deriving from the use of caregiver-informed measures to assess dog behavior, should be

considered when considering the generalizability of the findings. Third, the validity of translated versions of some scales (e.g., TDRS) requires further verification given the suboptimal fit found in the CFA. Future research could adopt a longitudinal design and include direct behavioral assessments to address some of these issues. A further limitation concerns the level of granularity of some MDQ response formats. Behavioral changes in the surviving dog were assessed by asking caregivers to indicate whether each behavior had increased, decreased, remained unchanged, or was unknown, but the magnitude of the reported change was not quantified. This format does not allow small changes to be distinguished from more marked behavioral alterations. Similarly, the MDQ assessed time since death using three broad categories: less than 6 months, 6–12 months, and more than 12 months. Consequently, the exact maximum time elapsed since the loss and the full range of this variable could not be determined. These response formats were consistent with the original exploratory design of the MDQ, which was conceived as a broad screening instrument for an under-investigated and emotionally sensitive topic. Because the questionnaire was already extensive and psychologically demanding, requesting highly specific dates, detailed temporal reconstructions, or precise estimates of behavioral change magnitude could have introduced additional uncertainty, especially for events for which memory may already have been imperfect.

Finally, the temporal distance of data collection should also be considered. Data were collected in 2018; therefore, the findings may not fully reflect current societal attitudes, veterinary end-of-life communication practices, or cultural norms surrounding pet loss and euthanasia. However, the purpose of the present study was to investigate whether specific death-related circumstances were associated with caregiver psychological responses and reported behavioral changes in surviving dogs. Therefore, the findings should not be interpreted as an assessment of current veterinary end-of-life practices, current pet ownership patterns, or how attitudes toward euthanasia may have changed over time in the two national populations. Rather, they should be interpreted as associations observed within the sampled population, which indicate what can happen.

Taken together, these limitations highlight the need for prospective or longitudinal studies beginning as close as possible to the time of loss, incorporating direct behavioral observations and more fine-grained measures of behavioral change magnitude, such as those previously used to assess surviving animals' responses to companion loss (e.g., Walker et al., 2016; Greene and Vonk, 2024), to better characterize the intensity and temporal dynamics of caregiver grief and surviving dogs' post-loss responses.

Conclusions

Our study indicates that euthanasia and expected death did not directly predict behavioral changes in the surviving dog but were indirectly associated through the caregiver's grief and emotional responses. These findings highlight the central role of caregiver psychological states in the post-loss environment, with implications for both human and animal well-being. Supporting the caregiver may therefore be considered part of a broader strategy to facilitate the surviving dog's adjustment to loss. Cultural context also played an important role. Differences between countries in emotional expression, attitudes toward euthanasia, and views of the human–animal bond suggest that grief and its impact on surviving pets are embedded within wider cultural frameworks. This underscores the need for culturally informed approaches to end-of-life communication and bereavement support in veterinary practice. Relational factors such as the bond between dogs, shared resources, and cohabitation length were key predictors of behavioral changes in the surviving dog, reinforcing the role of companionship and disruption of daily routines. While the concept of euthanasia is unlikely to be accessible to dogs, they may respond to indirect cues (e.g., prolonged illness, changes in interactions), and the possibility of some level of experiential or conceptual sensitivity to death cannot be entirely excluded. Because of the correlational design, these outcomes must be interpreted with caution. Future longitudinal and mixed-methods studies are needed to clarify causal pathways and mediating variables, such as caregiver attachment, previous experiences of loss, and the quality of veterinary communication, that influence how end-of-life decisions affect bereavement. By integrating psychological, relational, cultural, and environmental dimensions, more effective and sensitive support systems can be developed for both grieving caregivers and the behavioral and emotional needs of surviving dogs.

CRediT authorship contribution statement

Stefania Uccheddu: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Lucia Ronconi:** Writing – review & editing, Validation, Software, Formal analysis. **Mariangela Albertini:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Stanley Coren:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Goncalo Da Graça Pereira:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Anouck Haverbeke:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Daniel Simon Mills:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Ludovica Pierantoni:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Stefanie Riemer:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Ines Testoni:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Federica Pirrone:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Ethical statement

The study received ethical approval from the University of Padova Ethics Committee (46DF1164A03D63129CEE38D7571F8FB7). Informed consent was obtained from all participating caregivers prior to data collection. Participants were assured of the voluntary nature of their involvement, their right to withdraw at any time, and the anonymity and confidentiality of their responses.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT-4o to improve the readability of the manuscript during its preparation. The authors have reviewed and edited the content and take full responsibility for the final publication.

Funding

This research received no external funding.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jveb.2026.10.005.

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